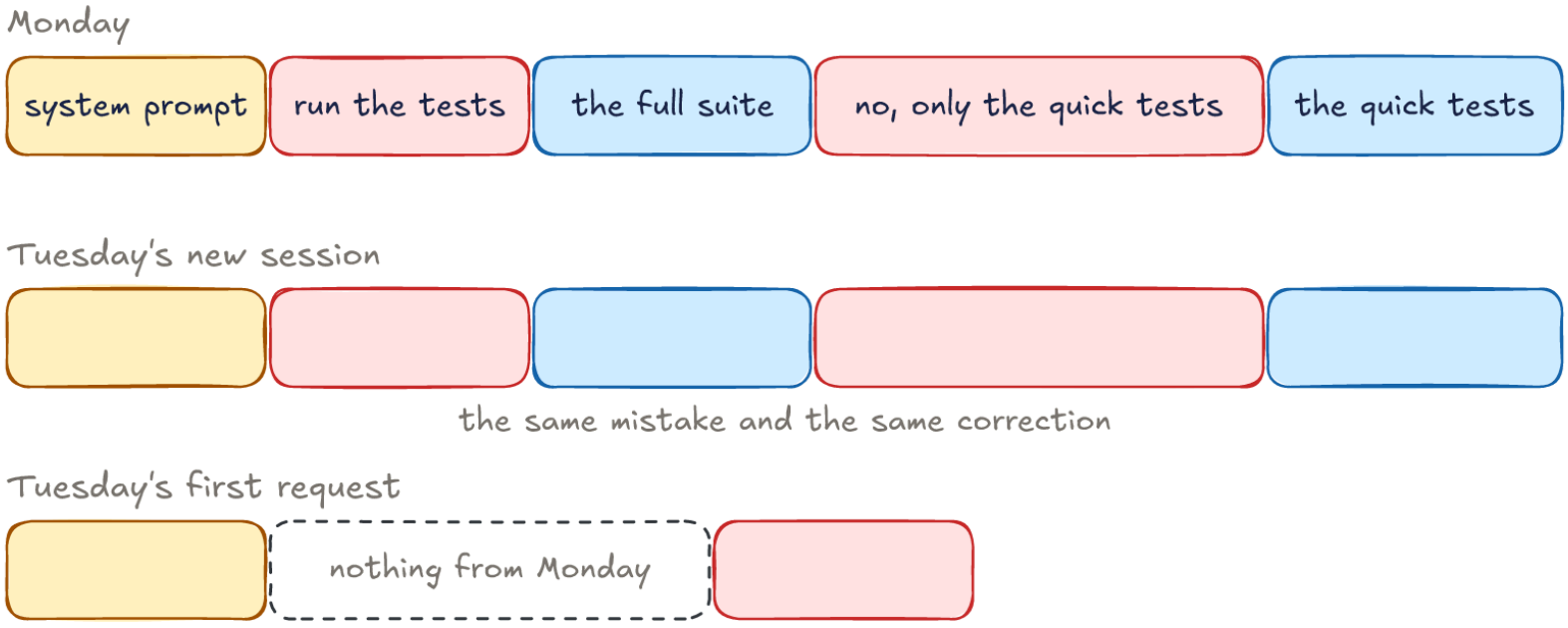
Agent Memory

EECS 498 AASE, Lecture 16

A correction the model forgets



How does the model keep what you taught it?

Today's schedule

- [x] Creating and storing memories
- [ ] Break
- [x] Loading and retrieving memories
- [ ] Questions

Part 1

Creating and storing memories

What to keep from a session

resend every session



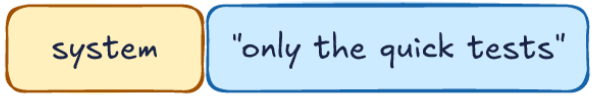
grows every session and is mostly files and failed tries

a summary of each session



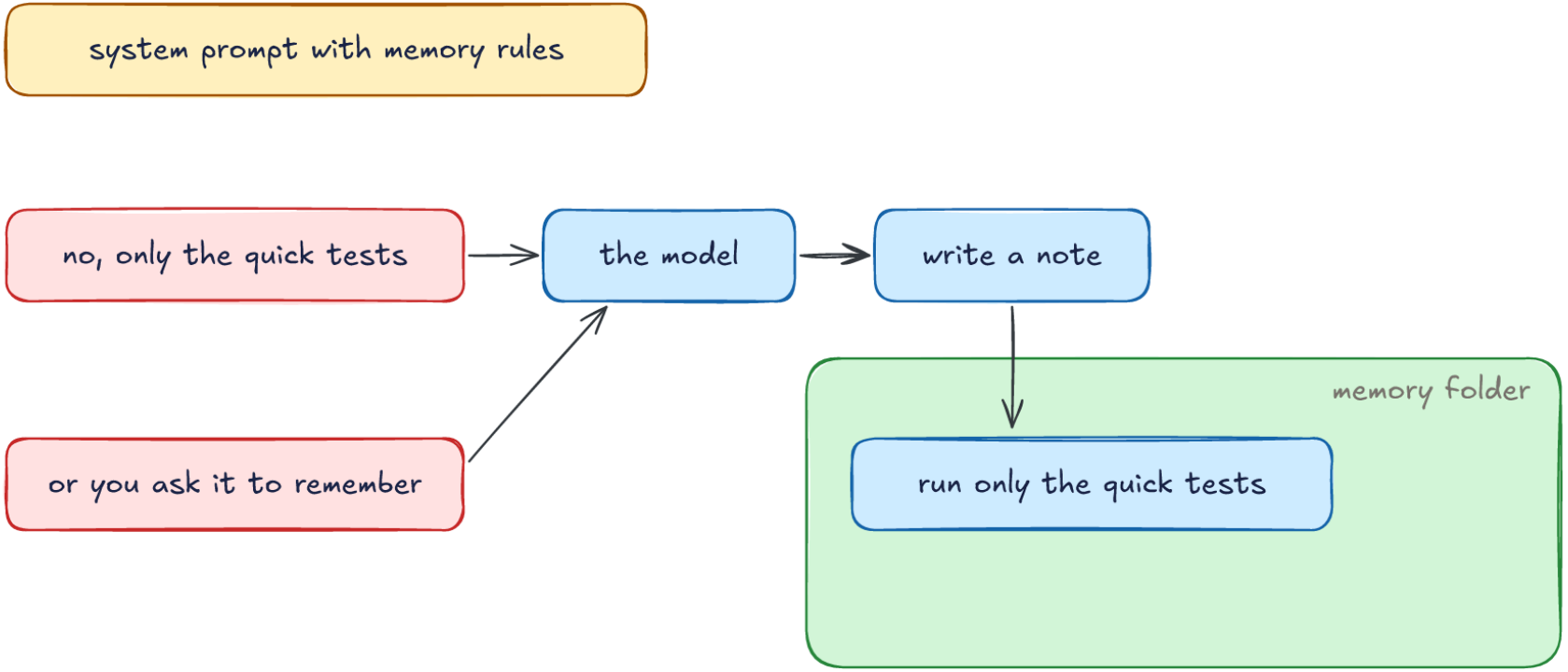
it is smaller but still grows every session

a note of what matters later

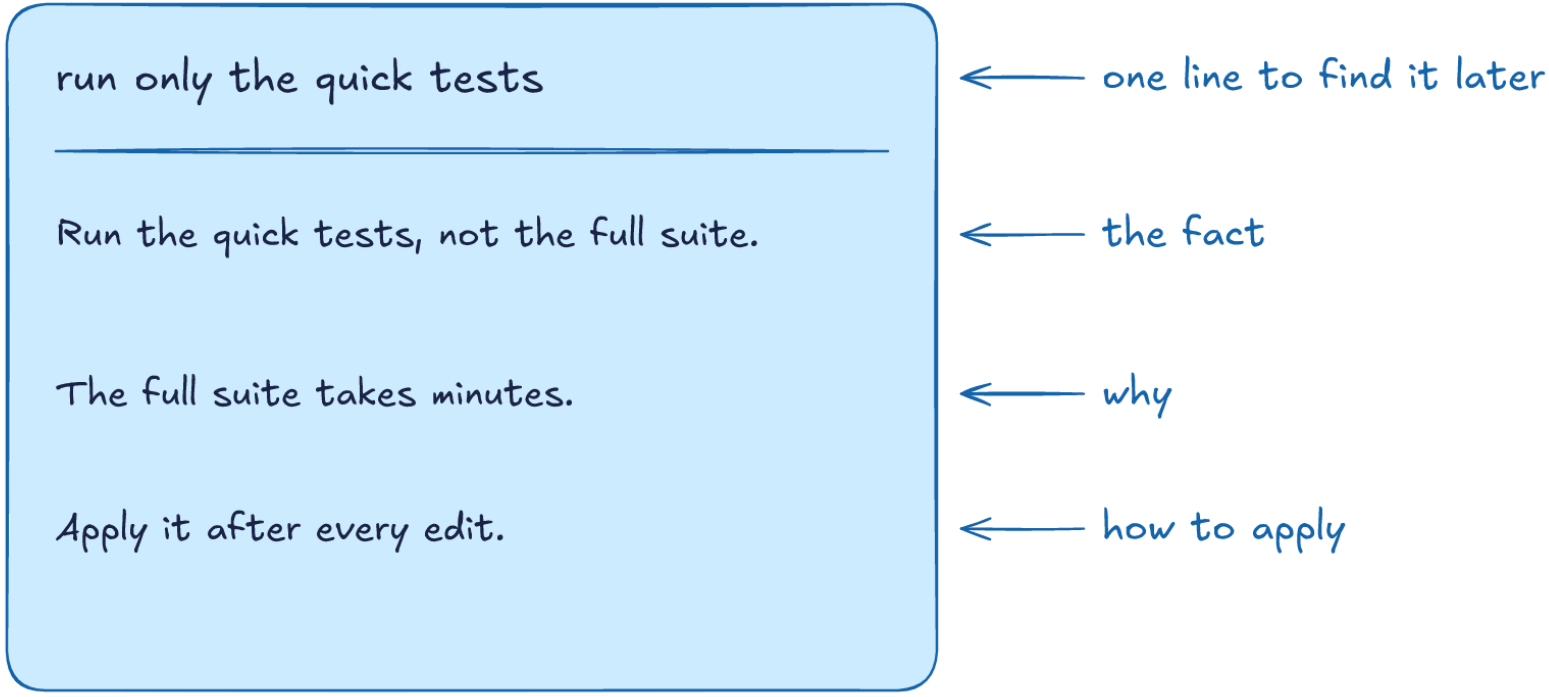


one sentence

A note the model writes

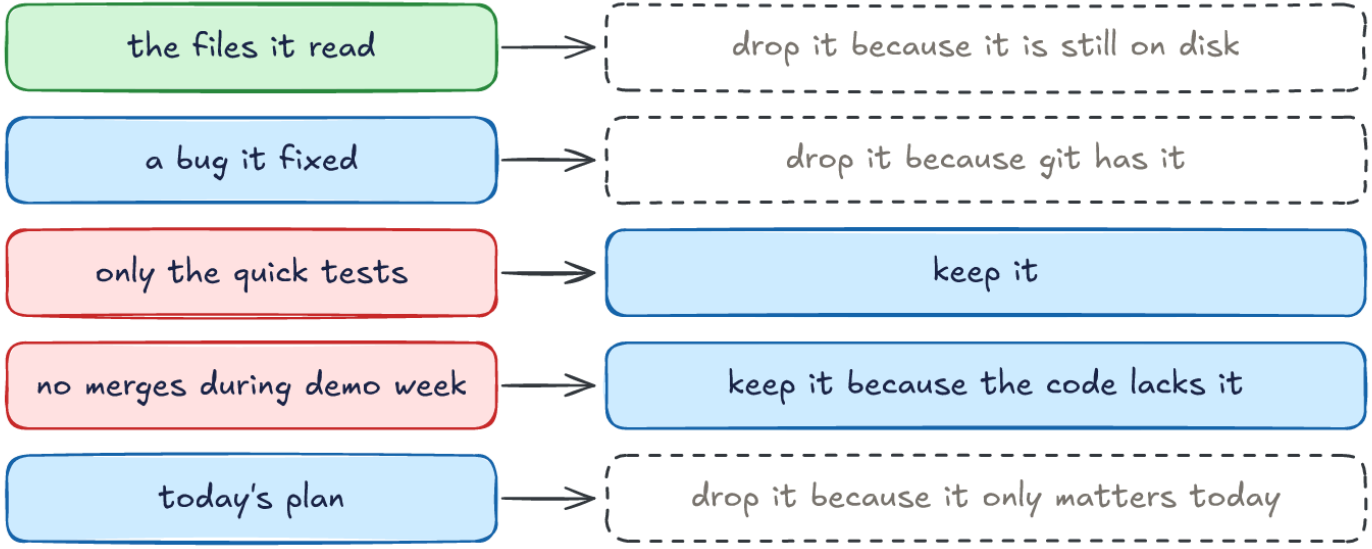


What a note holds



What is worth a note

Monday's session



keep what is true next week and not in the code

Four kinds of note

who you are

prefers short replies

how to work

only the quick tests

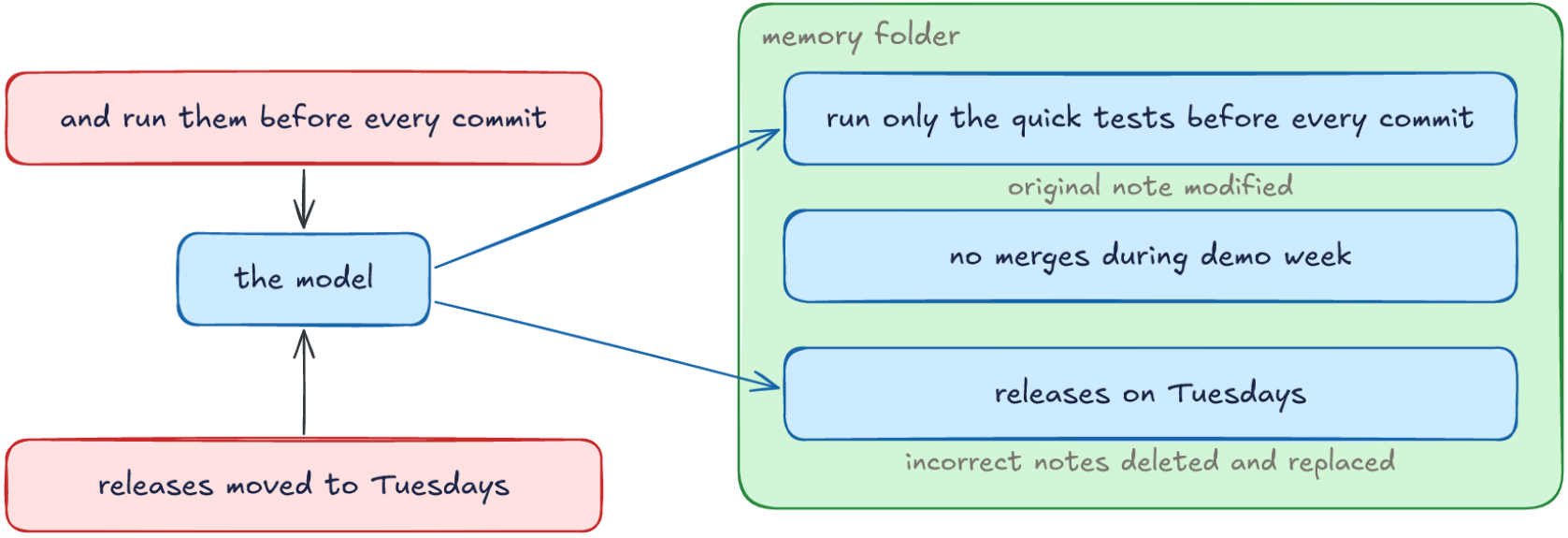
the project

no merges during demo week

where to look

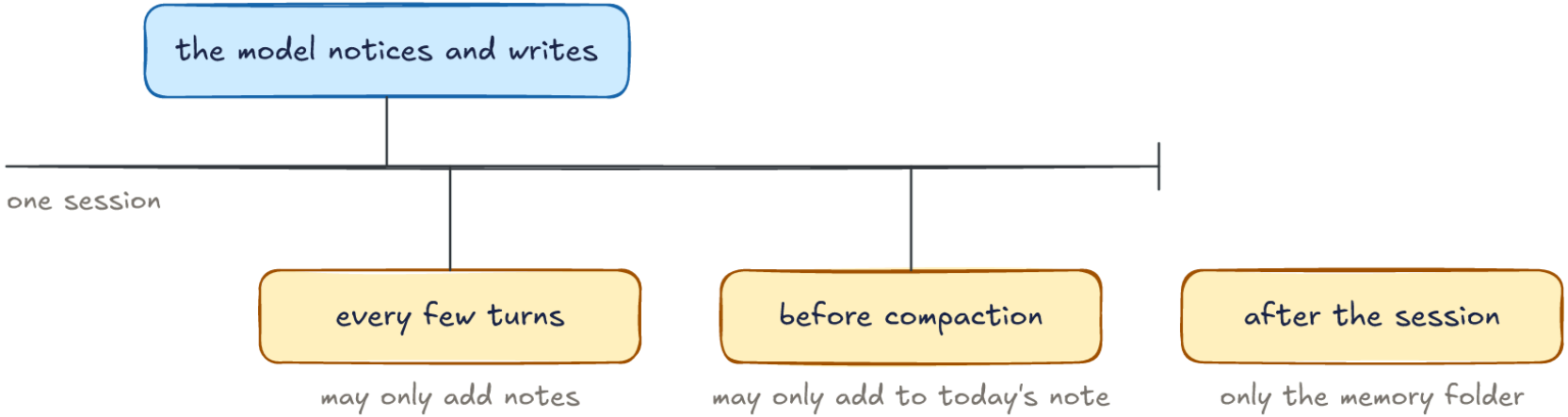
the dashboard that tracks bugs

Modifying notes



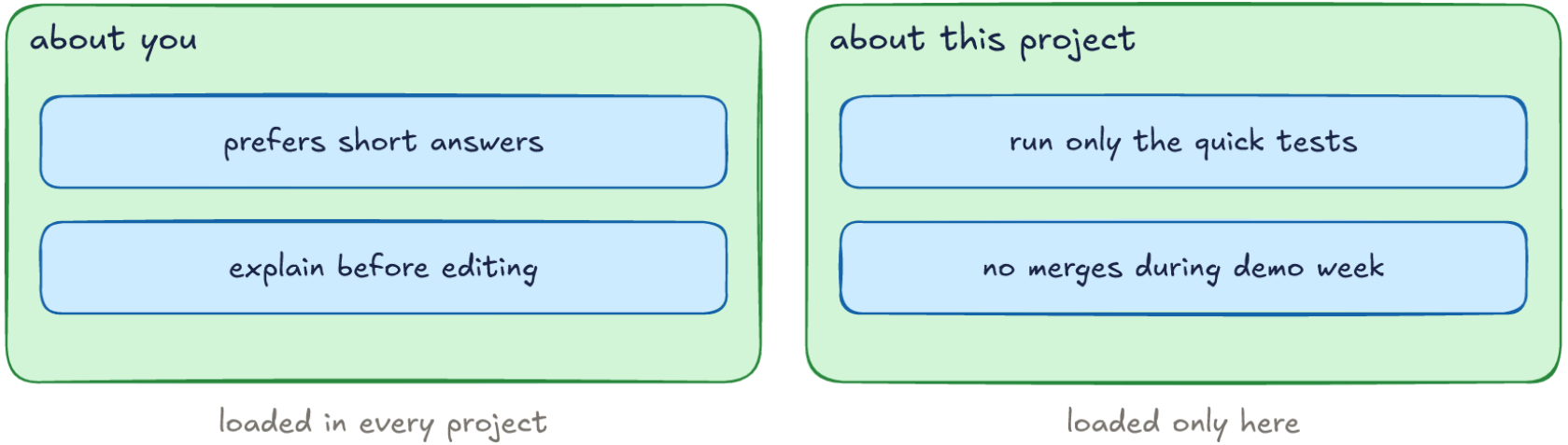
one note per fact, so two notes never disagree

When notes get written



less power when no one is watching

Notes about you and the project



plain files you can open, read and fix

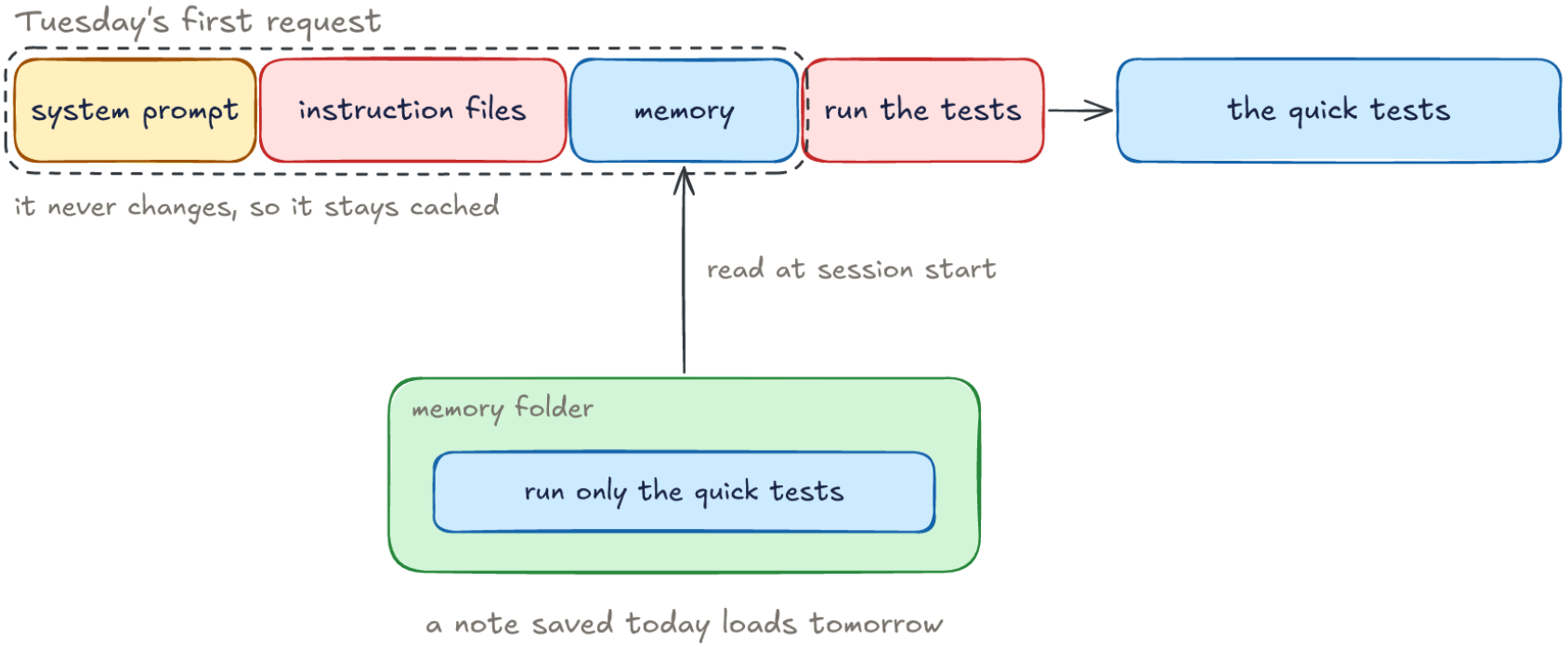
Chat break

Back in five minutes

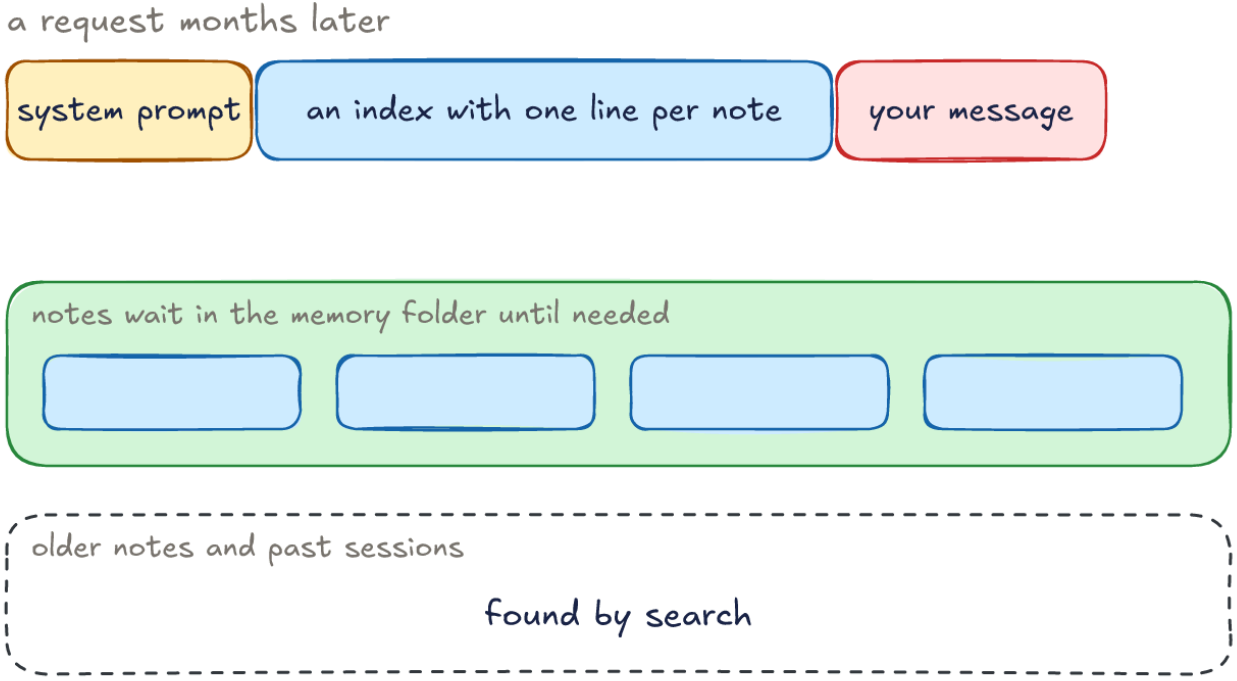
Part 2

Loading and retrieving memories

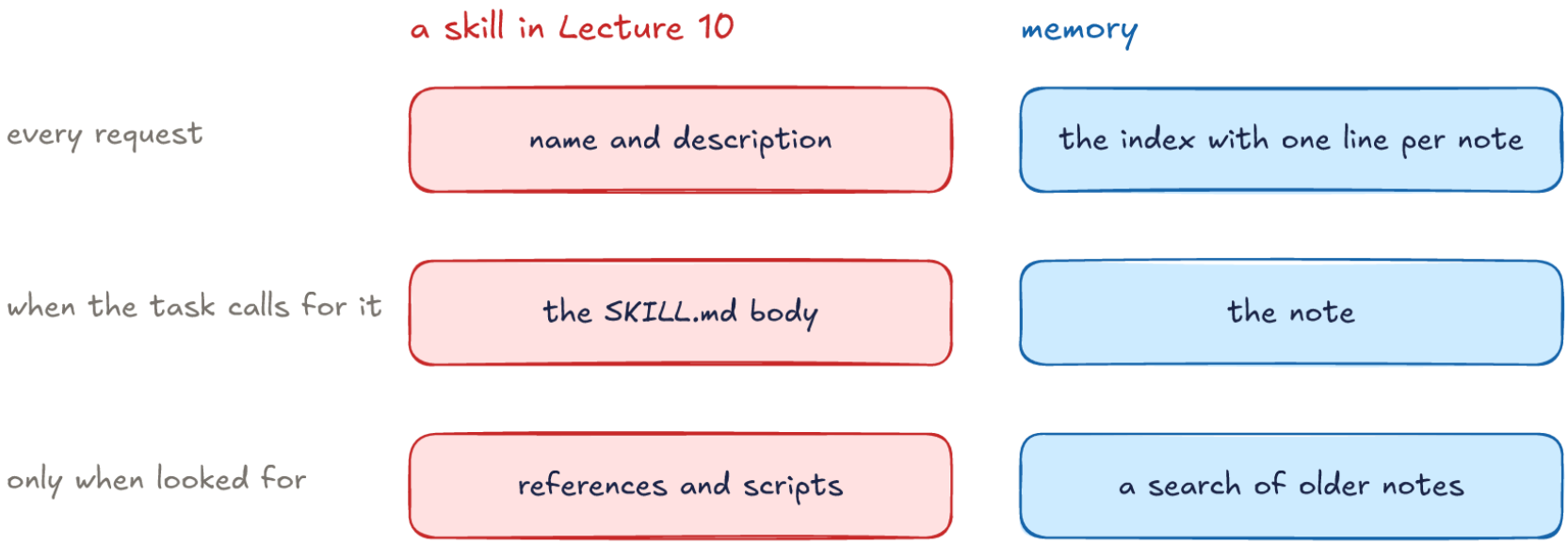
Loading notes back



Too many notes to load

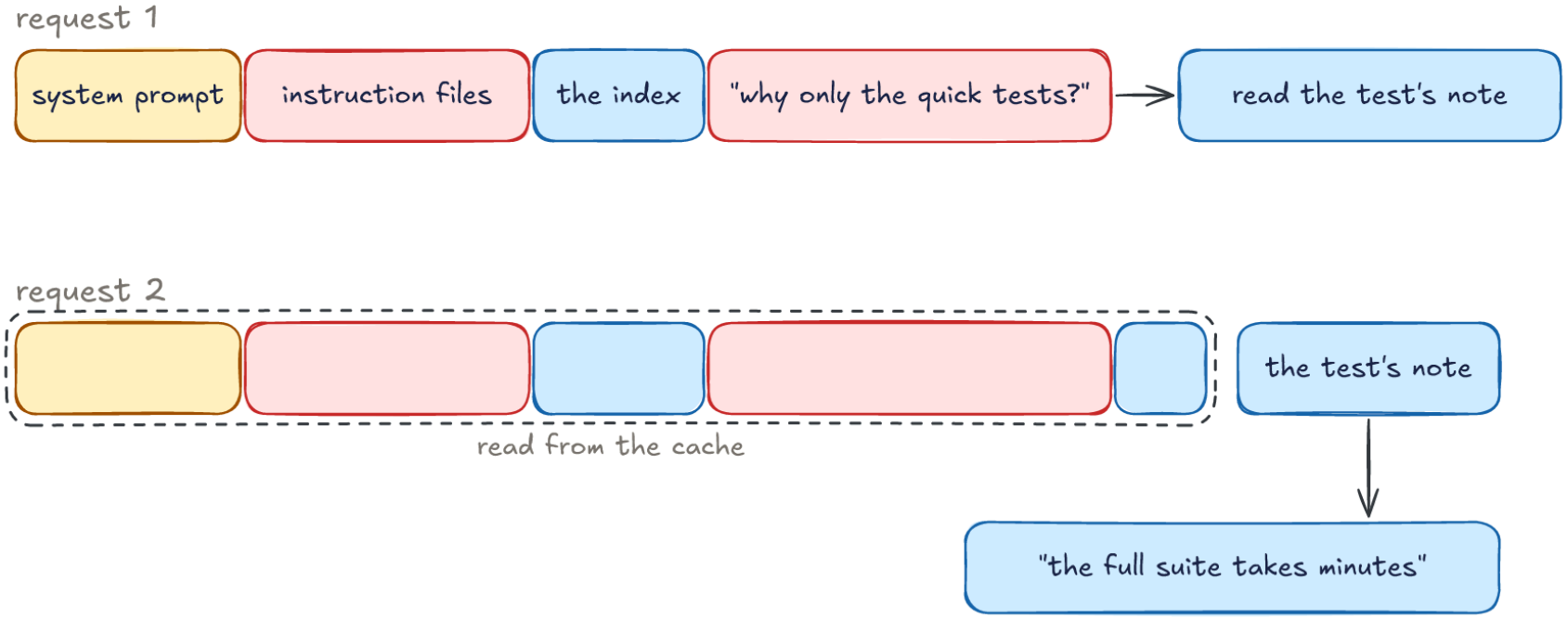


Loading in three levels

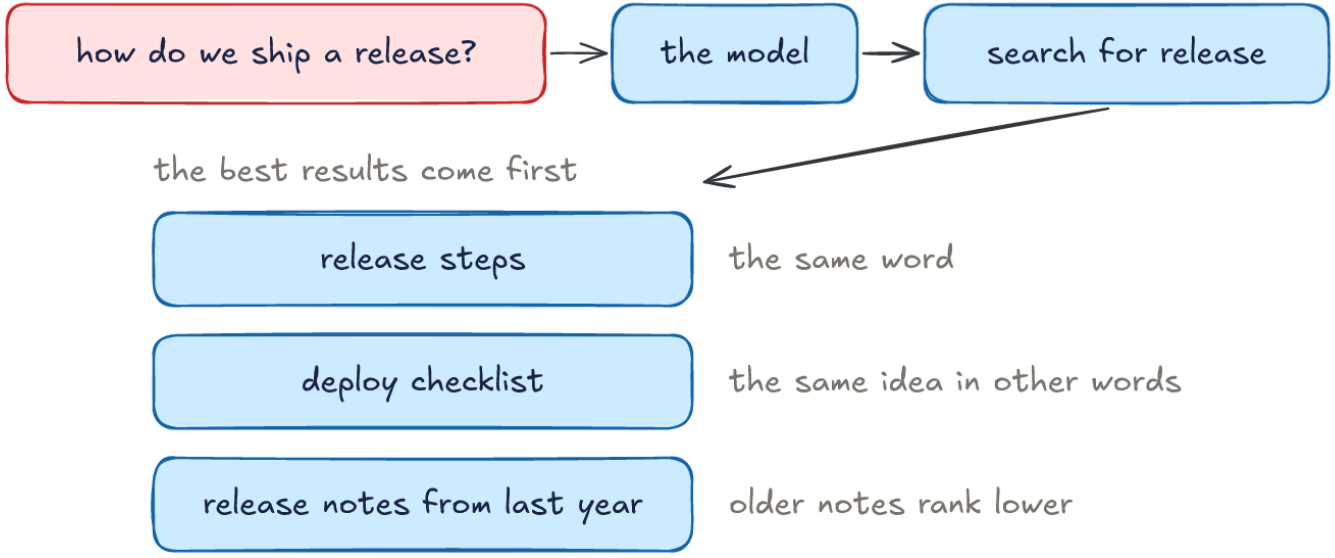


an index line is written like a skill description

Reading one note on demand

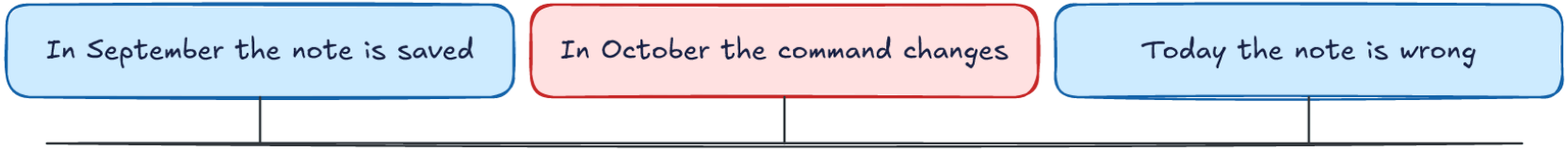


Searching older notes



the model reads the best one or two

A note that goes wrong

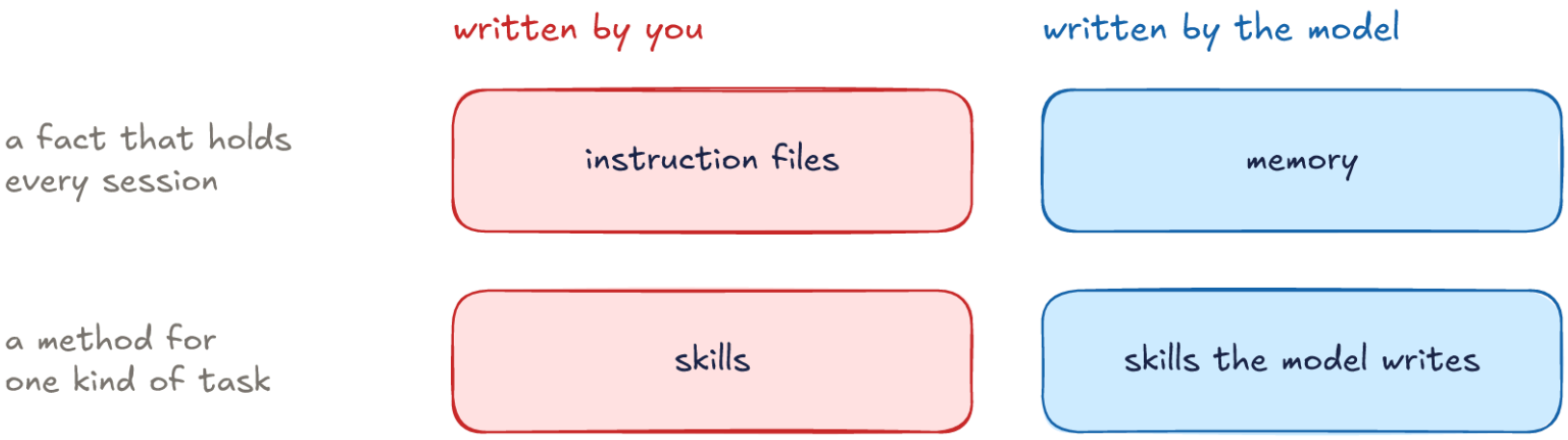


check it before acting on it

fix or delete the note

drop notes nobody uses

Facts and methods



all four are read index first

What to take away

- Memory is notes the model writes for later
- The harness also decides when the model writes
- Load one line per note and read the rest on demand
- Check a note before trusting it

This week

- Aider writes no memory of its own
- Standing rules go in a --read file

Sources

The Claude Code docs on memory
code.claude.com/docs/en/memory

The Claude API memory tool
platform.claude.com/docs/en/agents-and-tools/tool-use/memory-tool

Anthropic on effective context engineering
anthropic.com/engineering/effective-context-engineering-for-ai-agents

The MemGPT paper (2023)
arxiv.org/abs/2310.08560

More sources

Hermes Agent source
`github.com/NousResearch/hermes-agent`

OpenClaw source
`github.com/openclaw/openclaw`

Codex source for memories
`github.com/openai/codex`

Questions

AASE Lecture 16, Agent Memory